InfoMed — GovHealth-SLM: An Evidence-Grounded Small Language Model and High-Performance Retrieval Engine for Indian Public Healthcare Intelligence

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ABSTRACT

Public health data in India is published through independent government systems that measure different things in different ways: household surveys such as NFHS-5, monthly facility counts in HMIS, and vital-statistics surveys such as AHS. General-purpose large language models (LLMs) answer questions about these indicators fluently, but they can quote numbers that do not exist, confuse a State-wide rate with a district figure, or invent a time series. This paper presents InfoMed (GovHealth-SLM), a public healthcare assistant in which every stated number is traceable to an official dataset. It rests on a harmonised fact base of 1,180,313 records covering 778 canonical indicators for 28 States, 8 Union Territories and 700+ districts. Answers come from a three-tier pipeline: a hosted frontier model (NVIDIA Nemotron-3-Ultra-550B) screened by a numerical grounding validator, a Qwen3-1.7B small language model fine-tuned with 4-bit QLoRA, and a deterministic evidence provider that works without any model. A circuit breaker, clinical-safety abstention and clarification of ambiguous queries add reliability. A precomputed group-by index with array intersection cuts mean retrieval latency from 704.12 ms to 2.52 ms (279× faster). The system passed all 16 categories of a held-out benchmark, 11 of 11 live HTTP audit checks and 45 of 45 unit and grounding tests. InfoMed covers aggregate public data in English only and gives no diagnosis or treatment advice.

Keywords— *Public healthcare informatics, Small language model, QLoRA fine-tuning, Retrieval-augmented generation, Hallucination mitigation, Numerical grounding, NFHS-5, HMIS, FastAPI, React.*

I. INTRODUCTION

India's public health system produces a large volume of statistics, but they are scattered across portals built for different purposes. A researcher, journalist, district official or student who wants a simple figure, such as the share of institutional births in a district, must decide which system to consult, find the right indicator among hundreds, and work out whether the number comes from a household survey, a facility count or a registrar's survey.

Each source is authoritative within its own definition but they are not interchangeable. The National Family Health Survey (NFHS-5, 2019–21) samples households [1]; the Health Management Information System (HMIS) aggregates monthly facility reports [2]; the Annual Health Survey (AHS) provides vital-statistics rates [3]; and schemes such as PM-JAY add programme information [4]. Combining them carelessly yields answers that look precise and are wrong.

LLMs generate likely continuations and can produce fluent statements that no source supports, a behaviour known as hallucination [5], [6]. In medicine this is a documented hazard [7], [8]. For regional statistics it appears as non-existent indicators, State values reported as district values, and extrapolated trends for years with no data. We therefore design InfoMed so that **the language model is not the source of truth**: facts live in a verified dataset, retrieval selects the matching rows, and any generated sentence is checked against those rows before it reaches the user. The main contributions are:

- a harmonised fact base of 1,180,313 records and 778 indicators with source and period attached to every row;
- a three-tier answering architecture with a deterministic last tier that guarantees availability;
- an automated numerical grounding validator and a circuit-breaker for upstream failures;
- an exact-key inverted index that makes retrieval 279× faster than a linear scan; and
- a QLoRA-adapted Qwen3-1.7B domain model, evaluated with a 16-category benchmark, a live audit and automated tests.

II. PROBLEM STATEMENT

Public health data in India is fragmented across siloed portals with divergent recording methodologies, and commercial general-purpose LLMs show a high rate of numerical hallucination on regional Indian health queries. An assistant for this domain must:

1. unify the principal government sources into one verified fact base;

2. answer on that base with every numerical statement verified against the official data;
3. respond fast enough for interactive use over more than a million records; and
4. keep working when cloud model services fail.

How can a conversational system answer district-level Indian public-health questions in plain language while guaranteeing that every number it states can be traced to an official source, and still stay fast and available?

III. OBJECTIVES

1. To harmonise NFHS-5, HMIS, AHS and the National Hospital Directory into one fact base with a canonical indicator vocabulary and standard geography.
2. To design a three-tier answering architecture in which a hosted frontier model, a domain-adapted small model and a deterministic evidence provider back each other up.
3. To adapt Qwen3-1.7B to public health with 4-bit QLoRA on 10,000 instruction pairs.
4. To implement a numerical grounding validator that accepts a generated answer only when its numbers are supported by retrieved evidence.
5. To accelerate retrieval over 1,180,313 records from a second-scale scan to a few milliseconds.
6. To add clinical-safety abstention, clarification of ambiguous questions and a circuit breaker.
7. To evaluate the system with a held-out benchmark, a live audit and test suites, and present it through an interactive web interface.

IV. LITERATURE REVIEW

A. Large Language Models in Healthcare

Singhal *et al.* [9] introduced MultiMedQA and showed that instruction prompt tuning (Med-PaLM) improved answers to medical questions, while gaps with clinicians remained. BioGPT [10] is pre-trained on biomedical literature and ClinicalGPT [11] is fine-tuned on records and dialogue. These systems target clinical knowledge and biomedical text, not "what was indicator X in district Y in survey Z", whose correct answer lives in a statistical table. Reliability is a concern in its own right: Med-HALT [8] tests reasoning and memory hallucination, and SelfCheckGPT [12] flags unsupported statements by sample consistency, which judges consistency rather than correctness against a source.

B. Retrieval-Augmented Generation and Indexing

Retrieval-augmented generation (RAG) lets a generator draw on external evidence [13], [14]. Most work targets unstructured text with dense vectors [15] and approximate nearest-neighbour search [16]. Health indicators, however, are structured records addressed by exact keys (State, district, indicator, source), for which an exact index returns precisely the matching rows with no similarity threshold and no risk of a near-miss indicator. InfoMed therefore uses precomputed group indices rather than vector search.

C. Parameter-Efficient Fine-Tuning

LoRA freezes pre-trained weights and trains small low-rank matrices [17]; QLoRA back-propagates through a frozen 4-bit NormalFloat (NF4) model into the LoRA matrices, enabling fine-tuning on a single modest GPU [18]. The PEFT and Transformers libraries implement these methods [19], [20], and Qwen3 provides open models whose small members suit low-cost hosting [21].

D. Research Gaps

Three gaps emerge. (i) Medical LLMs are not built or evaluated for district-level Indian statistics. (ii) Hallucination work mostly diagnoses or detects errors, whereas a public-statistics assistant needs a design in which a number cannot reach the user unless verified. (iii) RAG research centres on text and similarity search and says little about serving very large structured tables interactively. Table I summarises how InfoMed responds.

TABLE I
SUMMARY OF LITERATURE SURVEY

Work	Technique	Limitation	InfoMed advantage
Singhal [9]	Prompt-tuned large LLM	No regional statistics	Verified Indian indicator data
Luo [10]	Biomedical pre-training	Literature-oriented	Survey and facility fact base
Wang [11]	Record/dialogue fine-tuning	No numeric check	Grounding validator
Pal [8]	Hallucination test suite	Diagnoses, no fix	Deterministic tier, abstention
Lewis [13]	RAG over text	Fuzzy for exact keys	Exact-key indexed retrieval
Dettmers [18]	4-bit QLoRA	No evidence layer	Qwen3-1.7B adapted to health

V. PROPOSED SYSTEM AND ARCHITECTURE

A. Overall Architecture

A user asks a question in the React front end, which calls the FastAPI back end [22], [23]. The back end first analyses the query: it intercepts diagnosis or prescription requests, detects ambiguity, and resolves synonyms and locations into filters. The retrieval engine then selects matching rows from the fact base, the evidence is passed to an answer tier, and the reply is returned as plain prose with source and period, together with table and chart views of the underlying rows (Fig. 1).

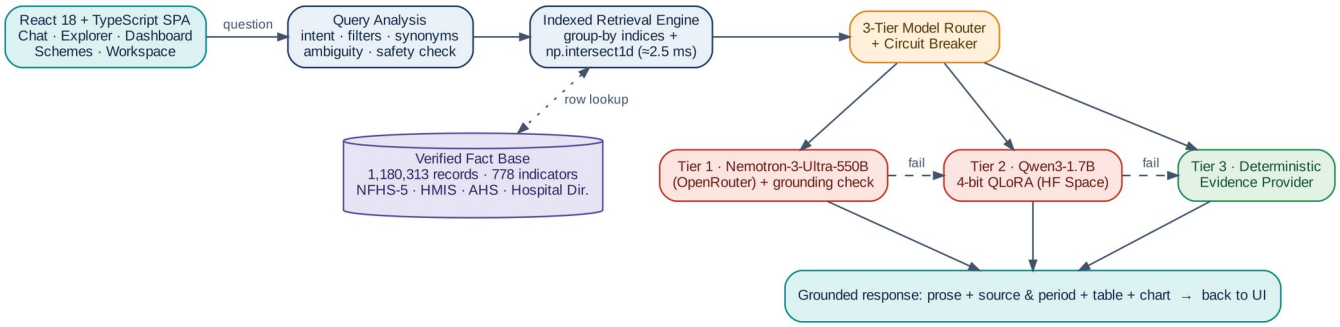


Fig. 1. Three-tier architecture of InfoMed: presentation layer, FastAPI back end, verified fact base and the three answer tiers with fail-over.

The three tiers back each other up (Table II). Tier 1 is NVIDIA Nemotron-3-Ultra-550B via OpenRouter [24], accepted only if the grounding validator confirms it. Tier 2 is Qwen3-1.7B fine-tuned with QLoRA. Tier 3 is the *DeterministicEvidenceProvider*, which performs verified synthesis and unweighted aggregation directly on retrieved rows and needs no language model.

TABLE II
ROLES OF THE THREE ANSWERING TIERS

Tier	Component	Failure behaviour
1	Nemotron-3-Ultra-550B (OpenRouter), checked by grounding validator	Rejected if numbers unsupported; skipped when circuit is open
2	Qwen3-1.7B, 4-bit QLoRA (HF Space / GPU)	Falls back to Tier 3
3	Deterministic evidence provider	Abstains when no evidence exists

B. Query Routing and Safeguards

Fig. 2 shows the routing flowchart. If no records match, the system abstains or asks for clarification; if the question is clinical, it declines and advises a Primary Health Centre (PHC). Otherwise it tries Tier 1 when the circuit is closed, falls to Tier 2 on failure or an ungrounded answer, and finally to Tier 3. Three guards surround the tiers: a **circuit breaker** (two consecutive failures, 60 s cool-down), **clinical-safety abstention**, and **ambiguity clarification** that asks the user to pick an indicator instead of guessing.

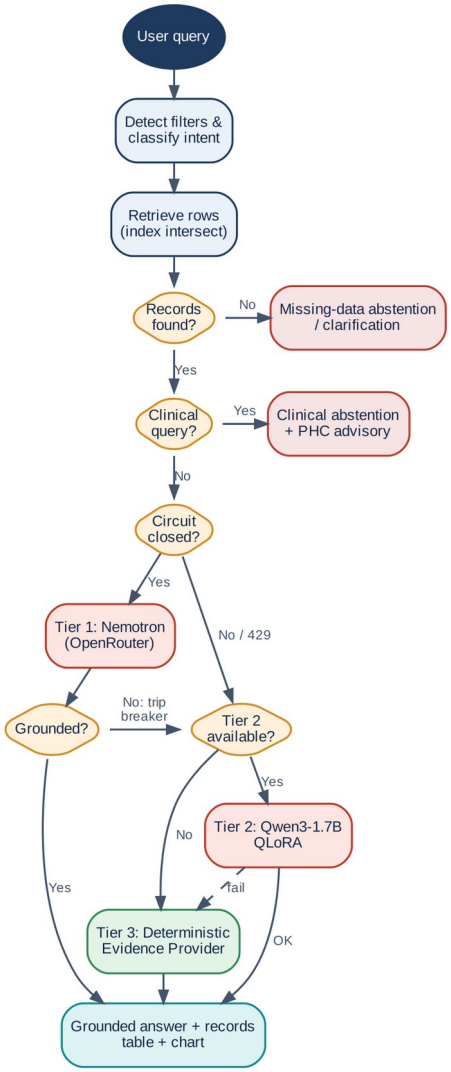


Fig. 2. Query-processing and routing flowchart with abstention paths and tier fail-over.

VI. DATASET ENGINEERING

The fact base, *healthcare_master_final_v3.csv*, holds 1,180,313 verified records for 778 canonical indicators across 28 States, 8 Union Territories and 700+ districts. Table III lists the four sources and Fig. 3 the harmonisation pipeline.

TABLE III
SOURCES INGESTED INTO THE MASTER FACT BASE

Source	Period	Publisher	Nature
NFHS-5	2019–2021	MoHFW, IIPS	Household sample survey

Source	Period	Publisher	Nature
HMIS	2019–2020	MoHFW	Monthly facility counts
AHS	2011–2014	Registrar General	Vital-statistics survey
Hospital Directory	Current	Govt. of India	Public PHCs, CHCs, hospitals

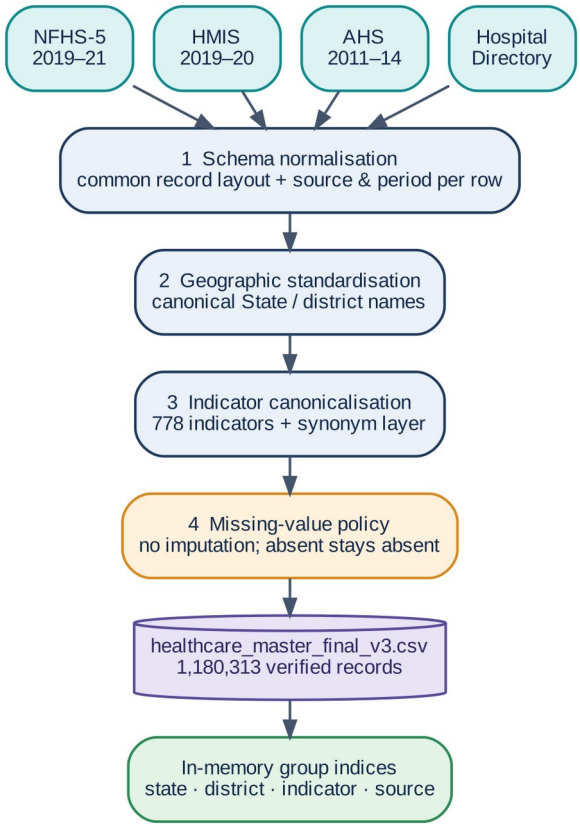


Fig. 3. Harmonisation of the four data sources into the master fact base and in-memory indices.

Schema normalisation. Sources differ in column names, units and layout, so each is converted to a common layout in which every row carries geography, a canonical indicator, a value, and its source and period. This is what lets the assistant attribute each figure (e.g., NFHS-5 to 2019–2021).

Geographic standardisation. State and district names are mapped to canonical spellings so the same place is recognised across sources. State-level and district-level rows stay distinct, which is essential for State-versus-district interpretation.

Indicator canonicalisation and missing values. Equivalent indicators are merged under 778 canonical names with a synonym layer (e.g., “c-section” resolves to caesarean deliveries). Missing values are never imputed: a place–indicator–period combination with no record stays absent and a question about it leads to explicit abstention rather than a plausible number.

VII. ALGORITHMS AND MATHEMATICAL FORMULATION

A. Fast Inverted Group-by Intersection

A linear scan testing every one of 1,180,313 rows with a string-contains filter took over 1,000 ms per query. The replacement builds once at start-up one dictionary per filterable field mapping each value to the sorted row positions that hold it (*state, district, indicator, source*). For field f and value v the entry is

$$I_f(v) = \{\, i \mid r_i[f] = v \,\} \tag{1}$$

For a query specifying the field set F, the candidates are the intersection of the selected entries, computed on unique sorted arrays (Fig. 4, Algorithm 1):

$$C = \bigcap_{f \in F} I_f(v_f) \tag{2}$$

A scan costs time growing with the row count N and the number of filters, while the indexed cost depends only on the sizes of the selected entries, usually far smaller than N:

$$T_{scan} = O(N \cdot |F|), \quad T_{idx} = O\Big(\sum_{f \in F} |I_f| \log |I_f|\Big) \tag{3}$$

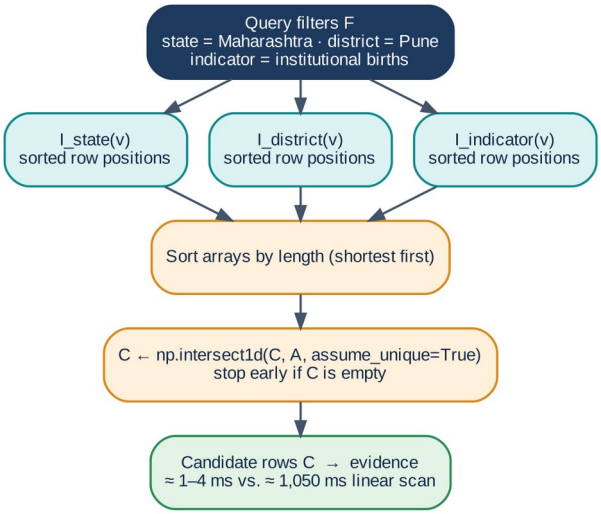


Fig. 4. Precomputed group indices feeding the array-intersection step.

Algorithm 1: Multi-filter array intersection

```
Input: filters F, values v_f, indices
Output: sorted candidate positions C
1 arrays <- []
2 for f in F with v_f given:
3   append index_f[v_f] (empty if absent)
4 if arrays empty: return all rows
5 sort arrays by length, shortest first
6 C <- arrays[0]
7 for A in arrays[1:]:
8   C <- intersect1d(C, A, assume_unique)
9   if C empty: break
10 return C
```

B. Automated Numerical Grounding Check

A fluent hosted-model answer is accepted only if every number in it traces to retrieved evidence. Let $N(a)$ be the numbers extracted from candidate answer a and E the numeric values in the retrieved rows (including Tier-3

derived values). The grounding score is the fraction of extracted numbers that match an evidence value within rounding tolerance ε, and the answer is accepted only when $g(a) = 1$ (Fig. 5):

$$g(a) = \frac{|\{x \in N(a) \mid \exists\, e \in E,\ |x - e| \le \varepsilon\}|}{|N(a)|} \tag{4}$$

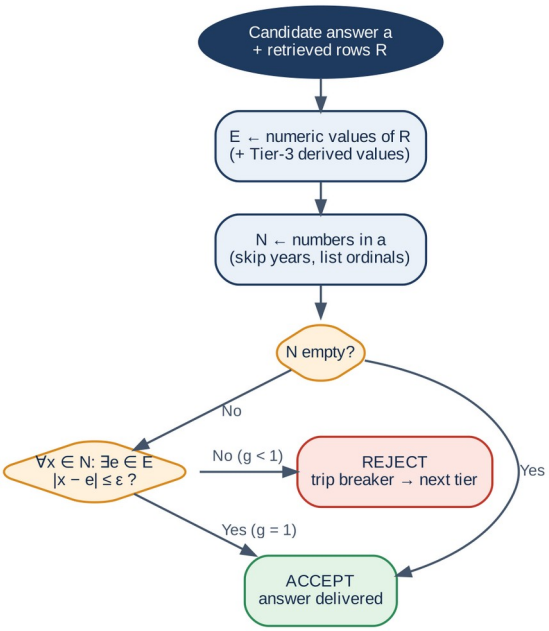


Fig. 5. Grounding validation flowchart (Algorithm 2). Years and list ordinals are excluded from the extracted numbers.

C. Circuit-Breaker State Machine

Hosted tiers can fail through network errors, quota limits or outages. A circuit breaker [25] stops the system from waiting on a service known to be failing (Fig. 6). In the *Closed* state calls pass and a counter c tracks consecutive failures; a success resets it. When c reaches 2 the breaker becomes *Open* and the next tier answers immediately. After a 60 s cool-down it becomes *Half-Open* and allows one trial: success closes the circuit, failure reopens it. Because Tier 3 needs no external service, a response can always be produced.

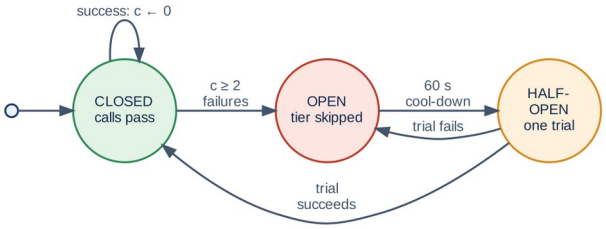


Fig. 6. Circuit-breaker state machine for hosted tiers.

D. 4-bit QLoRA Fine-Tuning of Qwen3-1.7B

Tier 2 adapts Qwen3-1.7B [21]. Base weights W_0 are stored in 4-bit NF4 and frozen; only the low-rank matrices B and A are trained [17], [18], so the adapted layer computes

$$W = W_0 + \frac{\alpha}{r} BA, \quad B \in \mathbb{R}^{d \times r},\ A \in \mathbb{R}^{r \times k} \tag{5}$$

With rank $r = 16$ and scaling $\alpha = 32$ the update is scaled by $\alpha/r = 2$. Fig. 7 shows the pipeline and Table IV the configuration. Training ran on a Google Colab Tesla T4 GPU, since the local 2 GB GPU was too small.

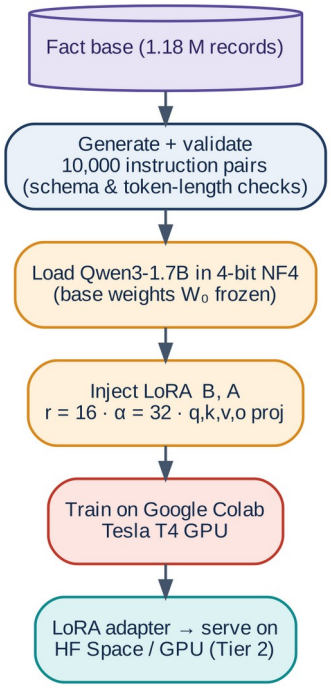


Fig. 7. QLoRA fine-tuning pipeline for the Tier-2 model.

TABLE IV
QLORA FINE-TUNING CONFIGURATION

Parameter	Value
Base model	Qwen/Qwen3-1.7B
Quantization	4-bit NF4
LoRA rank r / scaling α	16 / 32
Target modules	q_proj, k_proj, v_proj, o_proj
Training data	10,000 domain instruction pairs
Training environment	Google Colab, Tesla T4 GPU
Serving	Hugging Face Spaces / GPU back end

E. Evaluation Metrics

Retrieval latency is the mean time in milliseconds to return candidate rows; the acceleration factor is

$$S = \frac{T_{linear}}{T_{indexed}} \tag{6}$$

The *accuracy pass rate* is the share of benchmark items showing the expected, evidence-consistent behaviour:

$$PR = \frac{n_{pass}}{n_{total}} \times 100\% \tag{7}$$

Fact-grounding precision is the share of numerical statements in delivered answers that pass the check of (4), and the *failure recovery rate* is the share of induced upstream failures after which a valid grounded answer is still returned.

VIII. IMPLEMENTATION

Back end. FastAPI with pandas and NumPy [22], [26], [27] hosts the REST API, routing pipeline and index; PyTorch, Transformers and PEFT [28], [20], [19] are used

for fine-tuning. **Front end.** React 18, TypeScript, Vite, Tailwind CSS and Recharts [23]. Fig. 8 maps the five interface modules to the endpoints they use: the AI Chat Assistant (with an evidence drawer), the Data Explorer that pages and filters all 1.18 M rows, the Analytics Dashboard with heatmaps, the Schemes Catalog (PM-JAY, JSY, PMMVY) and the side-by-side Cross-Location Workspace.

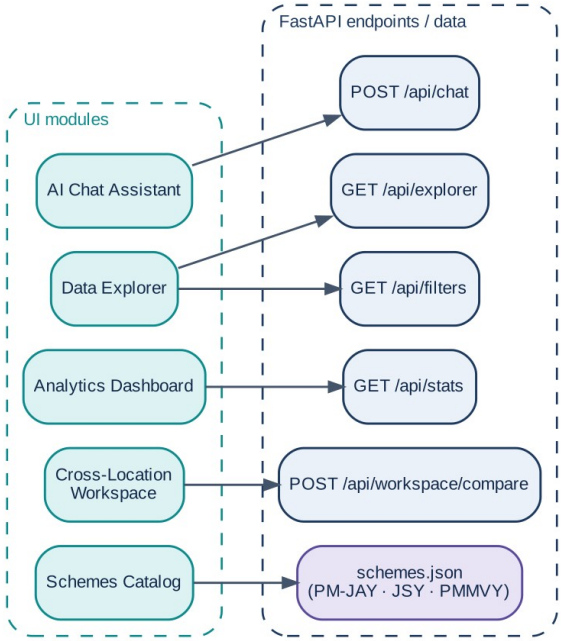


Fig. 8. Interface modules and the REST endpoints / data they use.

A *FormattedMessage* component removes raw markdown pipe tables from the chat text so replies read as prose, paired with interactive Table and Chart views; source and period are shown next to each figure. Development used a workstation with an NVIDIA GeForce MX450 (2 GB) and CPU PyTorch; QLoRA training ran on a Colab Tesla T4; hosted inference used OpenRouter (Tier 1) and Hugging Face Spaces / GPU (Tier 2). The source code is public [29].

IX. RESULTS AND DISCUSSION

A. Held-Out 16-Category Benchmark

The benchmark (*run_held_out_accuracy_audit.py*) probes correctness and restraint across 16 categories. The system passed **16 of 16 (100%)**; Table V lists the behaviour verified in each. Categories M–P test restraint: refusing to invent a 2026 figure, asking for clarification, surviving a cloud outage and declining clinical advice.

TABLE V
HELD-OUT BENCHMARK CATEGORIES AND RESULTS

ID	Category	Behaviour verified	Result
A	Exact lookups	Pune institutional births returned as 97.97% (NFHS-5)	PASS
B	Indicator discovery	Key Maharashtra indicators listed without locking to a district	PASS
C	State vs. district	Pune district distinguished from Maharashtra State	PASS
D	Synonym resolution	“c-section” resolved to caesarean deliveries	PASS
E	Source and period	NFHS-5 attributed to 2019–2021	PASS
F	Cross-location comparison	Pune compared with Nashik side by side	PASS
G	District rankings	Districts ordered by a health metric	PASS
H	Valid calculations	District mean given with unweighted-average caveat	PASS
I	Time-series trends	HMIS monthly deliveries trend	PASS
J	Follow-up summaries	Plain-language summary of session evidence	PASS
K	Multi-turn context	Indicator/State retained across drill-down	PASS
L	Multi-part questions	Institutional births and sex ratio answered together	PASS
M	Missing-data abstention	Year 2026 rejected without invented records	PASS
N	Clarification	Ambiguous comparison prompts for an indicator choice	PASS
O	Outage recovery	Seamless fallback to Tier 3 during cloud LLM failure	PASS
P	Clinical safety	Treatment query declined with PHC advisory	PASS

The benchmark is small (one or a few items per category), so a perfect score shows the designed behaviours work as intended but does not by itself prove equal performance on arbitrary questions.

B. Retrieval Acceleration

Table VI and Fig. 9 compare the linear scan with the indexed engine. Multi-filter lookups fell from about one second to a few milliseconds, and mean latency dropped from 704.12 ms to 2.52 ms, a **279×** acceleration by (6).

TABLE VI
RETRIEVAL LATENCY BEFORE AND AFTER INDEXING

Query type	Scan (ms)	Index (ms)	Speed-up
State + District	1,048.94	1.19	881.5×
State + Indicator	1,109.96	3.64	304.9×
District + Indicator	1,085.20	1.56	695.6×
Source + State	1,061.20	3.58	296.4×
Overall average	**704.12**	**2.52**	**279.4×**

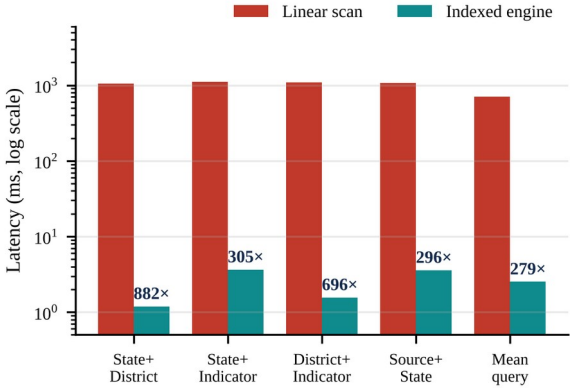


Fig. 9. Retrieval latency of the linear scan and the indexed engine (log scale); labels show speed-up.

C. Unit, Grounding and Live Audit Tests

Table VII summarises the automated suites. The live HTTP audit (*run_second_stage_audit.py*) exercised the running service and passed 11 of 11 checks; the unit and grounding tests passed 45 of 45, including grounding against the real dataset.

TABLE VII
SUMMARY OF AUTOMATED TEST RESULTS

Suite	Passed	Rate
Held-out 16-category benchmark	16 / 16	100.0%
Live HTTP audit	11 / 11	100.0%
Unit and grounding tests	45 / 45	100.0%

D. Comparison with Existing Approaches

Baselines were not run on this benchmark, so a numerical comparison of latency or hallucination rate would not be valid; Table VIII compares design features, based on published descriptions of the baselines and on the behaviour measured for InfoMed.

TABLE VIII
FEATURE COMPARISON WITH EXISTING APPROACHES

Feature	Hosted LLM	BioGPT [10]	InfoMed GovHealth-SLM
Answer source	Model weights unless retrieval is added	Biomedical literature in weights	Verified fact base of 1,180,313 records
Numerical validation	Not built in	Not built in	Grounding validator on every numerical answer
Indian district-level indicators	Not guaranteed	Not targeted	28 States, 8 UTs, 700+ districts
Retrieval latency	Not measured here	Not measured here	2.52 ms mean (indexed retrieval only)
Hallucination rate	Not measured here	Not measured here	No ungrounded number in 16 held-out categories
Offline readiness	Requires internet	Self-hostable	Tier 3 runs without any language model

The distinguishing feature of InfoMed is not model size but the combination of a verified fact base, validation of every number, and graceful degradation to a deterministic tier.

X. ADVANTAGES

First, every number is traceable to a source and period. Second, the deterministic tier gives 100% availability during outages, rate limits or quota exhaustion. Third, retrieval is interactive-speed over 1.18 M records. Fourth, a small fine-tuned model keeps hosting cost low. Fifth, clinical queries and unsupported periods are declined instead of answered. Finally, the modular design lets sources, tiers and interface modules evolve independently.

XI. LIMITATIONS

- Version 1 accepts English text only, limiting access for many intended users.
- Data reflects survey periods (NFHS-5 2019–21, HMIS 2019–20, AHS 2011–14), so recent changes are not captured.
- Only aggregate public data is used; no private health records or clinical advice.
- System-computed aggregates, such as district means, are unweighted and labelled as such.
- The benchmark is small and baselines were compared by features, not measured on the same questions.
- Tier 1 depends on a third-party hosted service, although the circuit breaker and lower tiers limit the impact.

XII. FUTURE SCOPE

- A multilingual voice interface using the Bhashini platform [30] so users can ask in Indian languages.
- Offline edge deployment of the small model and fact base on low-cost Android or Linux tablets for ASHA workers.
- Automated periodic ETL pipelines to synchronise national HMIS monthly data.
- A larger benchmark and a head-to-head measurement of hallucination rates against general-purpose models.

XIII. CONCLUSION

InfoMed is a public healthcare assistant whose numerical answers are tied to official government data. Its engineering contributions are a harmonised fact base of 1,180,313 records and 778 indicators; a three-tier architecture combining a hosted frontier model, a QLoRA-adapted Qwen3-1.7B and a deterministic evidence provider; a numerical grounding validator; an index that

cut mean query latency from 704.12 ms to 2.52 ms; and safeguards including a circuit breaker, clinical-safety abstention and clarification. It passed 16/16 benchmark categories, 11/11 live audit checks and 45/45 unit and grounding tests. The result lets officials, researchers and citizens query fragmented national health statistics in plain language with the source and period of each figure visible. Source code and evaluation scripts: https://github.com/Sush2307/InfoMed.

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